

Gmail MCP Multi-User Server

Project Report

Introduction

The Gmail MCP Multi-User Server is a Python-based application developed using FastAPI and FastMCP. It enables users to connect multiple Gmail accounts and perform email-related operations through REST APIs and the Model Context Protocol (MCP). The project securely stores Gmail account information and communicates with Gmail services using SMTP and IMAP protocols.

Objective

The objective of this project is to develop a server that allows multiple users to manage their Gmail accounts securely through a single application. It provides APIs for registering Gmail accounts, fetching emails, and managing account information while supporting AI applications through MCP.

Technologies Used

- Python
 - FastAPI
 - FastMCP
 - SQLite
 - SQLAlchemy
 - SMTP
 - IMAP
 - Cryptography (Fernet Encryption)
 - Uvicorn
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Project Structure

The project is organized into different modules to simplify development and maintenance.

- **app.py** – Starts the FastAPI application.
- **routes/** – Contains API endpoints.
- **services/** – Implements business logic.
- **database/** – Stores database models and operations.
- **gmail/** – Handles Gmail communication using SMTP and IMAP.
- **mcp_server/** – Provides MCP tools and resources.
- **utils/** – Contains helper functions and utilities.

Working of the Project

1. The FastAPI server is started using Uvicorn.
 2. A user registers a Gmail account through the **POST /accounts** API.
 3. The Gmail email address and App Password are securely stored in the database.
 4. The server connects to Gmail using SMTP for sending emails and IMAP for receiving emails.
 5. Users can retrieve their registered accounts using the **GET /accounts** endpoint.
 6. The **POST /accounts/{account_id}/fetch** endpoint manually checks the Gmail inbox and retrieves new messages.
 7. The **DELETE /accounts/{account_id}** endpoint removes a registered Gmail account from the database.
 8. The **GET /health** endpoint checks whether the application and database are running correctly.
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API Endpoints

Method	Endpoint	Description
GET	/health	Checks application and database status
POST	/accounts	Registers a new Gmail account
GET	/accounts	Retrieves registered Gmail accounts
DELETE	/accounts/{account_id}	Deletes a Gmail account
POST	/accounts/{account_id}/fetch	Fetches emails from Gmail

Conclusion

The Gmail MCP Multi-User Server provides a simple and organized way to connect multiple Gmail accounts using Python and FastAPI. It integrates Gmail services through SMTP and IMAP while exposing functionality through REST APIs and MCP. The project demonstrates secure account management and serves as a foundation for AI-powered email applications.